

Behind the Extra Layer: Multi-Factor Authentication's Role in Securing a Digital Future

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What is multi-factor authentication and its role in securing a digital future? With phishing attacks and data breaches on the rise, an extra layer of protection or multi-factor authentication (MFA), is increasingly essential across industries to securing online accounts. While cybersecurity experts agree that any type of MFA is better than no MFA, some methods are more secure while others are easier to implement. Since passwords are no longer an effective single authentication method, several types of MFA are available including one-time passcodes and push notifications, but physical passkeys remove the need for passwords and are becoming more mainstream as experts agree of their unmatched defense against hackers (Lyons, 2025; Mohammed et al., 2023).

Limitations of Passwords

Passwords are the most prevalent method of authentication to protect accounts and sensitive data. However, hackers are exploiting weak passwords to gain access to accounts using several attack techniques. Brute-force attacks involve a hacker repeatedly attempting to break into an account guessing at all variations or combinations of common passwords until one is correct (Bagane et al., 2025). Security researchers proved weak passwords are still being used exhibited in a situation regarding McDonald's AI hiring platform Paradox.ai. The researchers accessed approximately 64 million application records by guessing the username and password of "123456" and were able to access the administrative account. Even though the researchers reported the issue and Paradox.ai claimed ownership and vowed to increase security measures, if hackers were able to access the same account, contact information and employment history for any job candidate who has ever applied for work at McDonalds could have easily been exposed or sold (Greenberg, 2025). Encouraging users to implement strong and unique passwords can be beneficial, though with social engineering tactics, phishing attacks designed to trick unsuspecting users into providing their login credentials to fake websites, and difficulty in remembering several unique and complex credential combinations, relying solely on the use of passwords is no longer enough to fully secure accounts (Bagane et al., 2025; Mohammed et al., 2023).

Common Multi-Factor Authentication Methods

Multi-factor authentication plays a significant role in managing digital identities as cybersecurity threats such as phishing and password cracking attacks have increased the need for additional safeguards on user accounts. While there are many forms of verification, they fall into one of three categories: something you have (ownership), something you know (knowledge), and something you are (biometrics). Ownership encompasses objects like bank cards, physical security keys, and cell phones along with push notifications and one-time passcodes (OTP). The main shortcoming for this type of authentication is the risk of the object being lost, stolen, or broken. Knowledge includes passwords, pins, and answers to security questions making it the most common form of authentication but easy to forget. Biometrics includes physical characteristics like facial recognition or fingerprint scans and behavioral biometrics like keystroke pattern, typing speed, and signatures. Biometrics are more secure than other methods but can still be spoofed (Mohammed et al., 2023).

The most utilized and easiest to implement method of MFA is the use of a one-time passcode each time a user attempts to login distributed via email, SMS, or push notification. This method does not require additional information to activate but does take time waiting for

and entering the code into the platform. In terms of security, the passcodes can be intercepted and email phishing can encourage users to enter their credentials and OTPs into fake websites allowing hackers access to the account (Mohammed et al., 2023). The push notification method requires the user to download an authenticator app on their smartphone and invite attacks like MFA fatigue attacks where a hacker will overwhelm a user with authentication push requests until the user accepts out of annoyance or ambivalence. The simplicity of tapping a notification can easily be manipulated and requires the user to always have their smartphone accessible to access their account (Mohammed et al., 2023).

The Future of Multi-Factor Authentication

With the prevalence of phishing and AI-driven social engineering attacks, industry experts are considering if one-time passcodes and push notifications are outdated methods of MFA. Some organizations have started to move past these methods in favor of phishing-resistant physical passkeys. Instead of requiring an additional authentication method on top of a password, physical passkeys rely on a pin or biometric and ownership of the physical device as the two methods of authentication (Lyons, 2025; McConachie, 2025).

With the adoption of physical passkeys, there is no avenue for phishing as the security key requires the user to be present in order to plug in or tap the key to their device. The authentication method is FIDO2 which relies on asymmetrical cryptographic key pairs, one public key on the server and the other stored directly on the device (Lyons, 2025). This prevents the ability to share or intercept a one-time passcode or accidentally accept a push and eases the burden of remembering and entering several username and password combinations. The time it takes to log on to accounts decreases by 73% compared to the OTP method (Lyons, 2025). Hardware passkeys require initial investment in the device and to increase compatibility across services. However, if they become mainstream, cumbersome authentication requirements will be a thing of the past (Lyons, 2025; McConachie, 2025).

Conclusion

Adding an extra layer of protection to an important account can keep users more secure from potential data breaches and hackers trying to access valuable information. One-time passcodes and push notifications can be intercepted or phished by unsuspecting users and still allow hackers to gain access. Hardware passkeys are the strongest measure against password cracking, phishing, and social engineering as the user has to be physically present at their device to login. While any MFA provides more security than not using it at all, cybersecurity is always evolving and physical passkeys currently provide the strongest protection for users and organizations.

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